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DEPARTMENT OF MACHINE LEARNING

(UG Program: B.E. in Artificial Intelligence and Machine Learning)

Course : Social Network Analysis
Course Code: 24AM7PESNA

Link Prediction in a Co-authorship Network: Recommending Future Research Collaborations

Alternative Assessment Tool (AAT)

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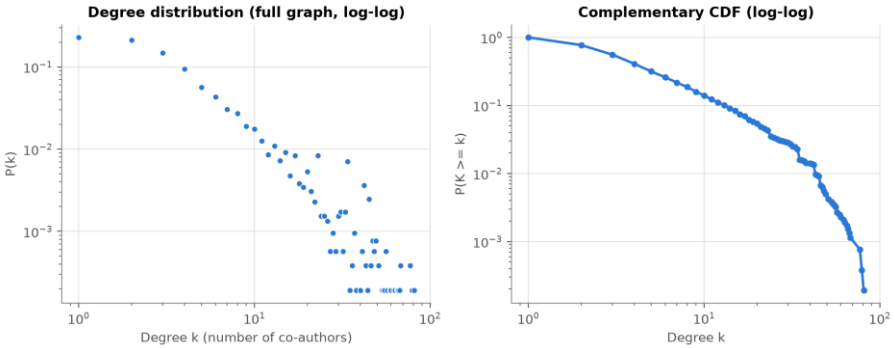
Problem Definition

- **Domain:** academic collaboration - who writes papers with whom.
- **Link prediction:** given today's co-authorship network, which unconnected pairs of authors are most likely to collaborate in the future?
- **Why it matters:** collaborator recommendation (Google Scholar, ResearchGate), team formation, reviewer suggestion, forecasting the growth of a field.
- **Formal task:** hide part of the edges, score every non-adjacent pair s(u, v) using only the visible graph, and rank the hidden edges at the top.

Network element	Mapping
Node	Author (researcher)
Edge	Co-authored >= 1 paper
Type	Undirected, unweighted
Self-loops	Removed
Experiments on	Largest connected component

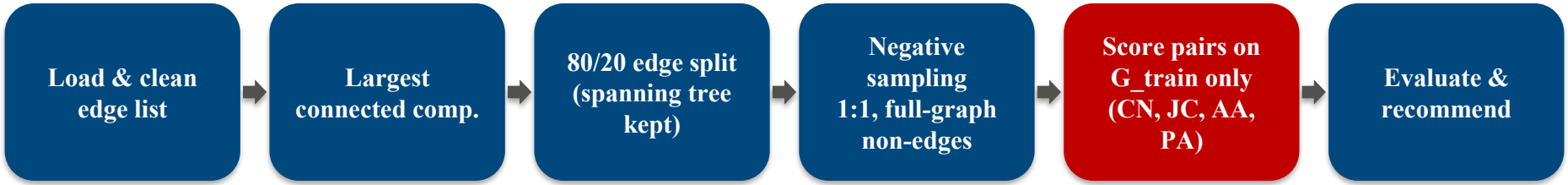
Dataset and Network

Statistic	Full graph	LCC
Nodes (authors)	5,242	4,158
Edges (co-authorships)	14,484	13,422
Density	0.00105	0.00155
Average degree	5.53	6.46
Avg. clustering coefficient	0.530	0.557
Transitivity	0.630	0.629
Connected components	355	1
Diameter / avg. path length	-	17 / 6.05



- SNAP ca-GrQc (arXiv General Relativity, 1993-2003); 12 self-loops and 14,484 duplicate lines removed.
- Sparse but highly clustered and small-world; heavy-tailed degrees (median 3, max 81).
- Louvain: 42 communities, modularity 0.846.

Methodology



- **Four heuristics written from scratch** with Python neighbour sets (src/scores.py) and cross-checked against NetworkX on 16,106 pairs: max difference 0.
- **Leakage-free:** test edges are removed before any score is computed; train positives are scored with their own edge hidden.
- **Evaluation:** ROC-AUC, Average Precision, Precision@K, P/R/F1 at a train-chosen threshold, 5-seed stability, hard negatives, hit rate.
- **Recommendation:** rank all distance-2 non-adjacent pairs; explain each by its shared co-authors.

Heuristic Formulas

Common Neighbours

$$CN(u, v) = |\Gamma(u) \cap \Gamma(v)|$$

Number of shared co-authors (triadic closure).

Jaccard Coefficient

$$JC(u, v) = \frac{|\Gamma(u) \cap \Gamma(v)|}{|\Gamma(u) \cup \Gamma(v)|}$$

Shared co-authors as a fraction of all co-authors.

Adamic-Adar

$$AA(u, v) = \sum_{w \in \Gamma(u) \cap \Gamma(v)} \frac{1}{\log k(w)}$$

Shared co-authors weighted by 1/log(degree): rare collaborators count more than hubs.

Preferential Attachment

$$PA(u, v) = k(u) \cdot k(v)$$

Product of degrees: 'rich get richer', favours hubs.

Experimental Setup

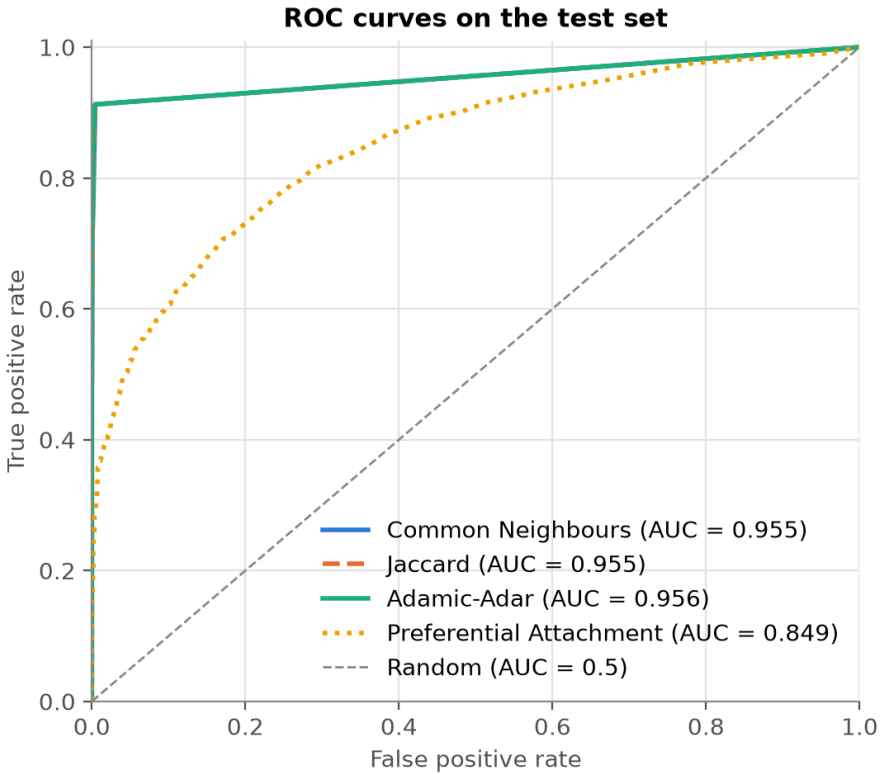
- **Split (seed 42):** 10,738 train edges / 2,684 test edges (20 % of 13,422); a random spanning tree of 4,157 edges always stays in train - the training graph keeps all 4,158 nodes and stays connected.
- **Negative sampling:** pairs that are NOT edges of the full graph, 1:1 - 10,738 train and 2,684 test negatives (disjoint).
- **Leakage avoided:** every score is computed on the training graph only.
- **Threshold:** the value that maximises F1 on the TRAINING pairs, then applied unchanged to the test pairs.
- **Metrics:** ROC-AUC, Average Precision, Precision@K (K = 10, 20, 50, 100), Precision/Recall/F1.
- **Robustness:** 5 random splits (seeds 42, 43, 44, 45, 46); hard negatives = 2,684 non-links at distance 2.

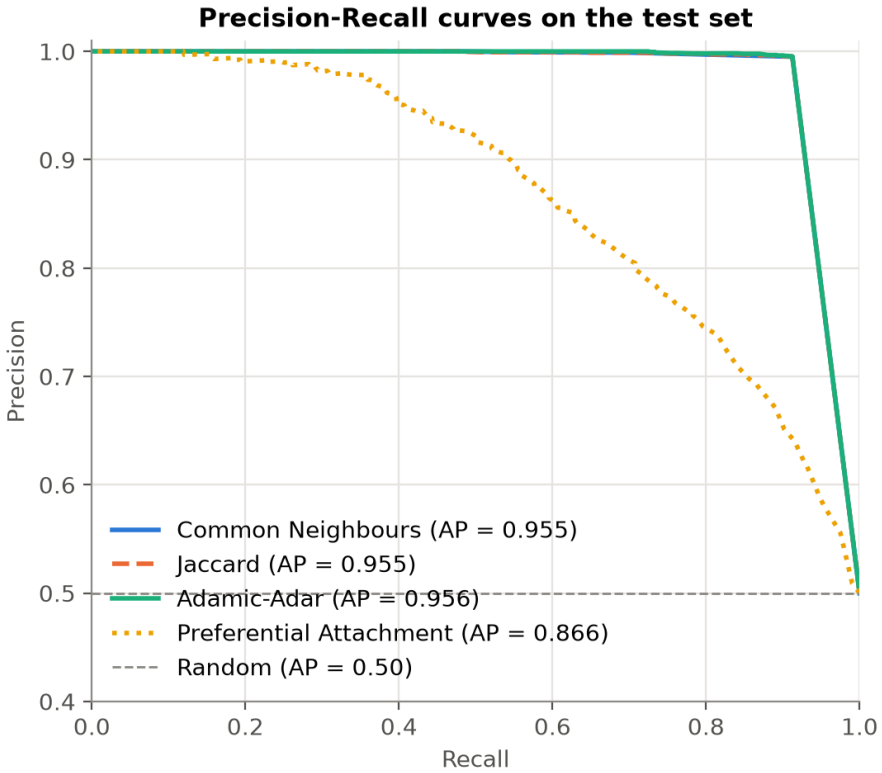
Results

Method	ROC-AUC	AP	Threshold	Precision	Recall	F1	P@100
Common Neighbours	0.9554	0.9550	1	0.995	0.912	0.952	1.000
Jaccard	0.9555	0.9555	0.0119	0.995	0.912	0.952	1.000
Adamic-Adar	0.9557	0.9558	0.2423	0.995	0.912	0.952	1.000
Preferential Attachment	0.8490	0.8662	10	0.652	0.902	0.756	1.000

- Local indices (CN, Jaccard, AA) nearly tied: ROC-AUC 0.9554-0.9557; Preferential Attachment far behind (0.8490).
- 8.8% of test positives but 99.6% of random negatives share no co-author - 'at least one shared co-author' already separates most pairs.
- Best overall: Adamic-Adar - mean ROC-AUC 0.9568 +/- 0.0028 over 5 splits.

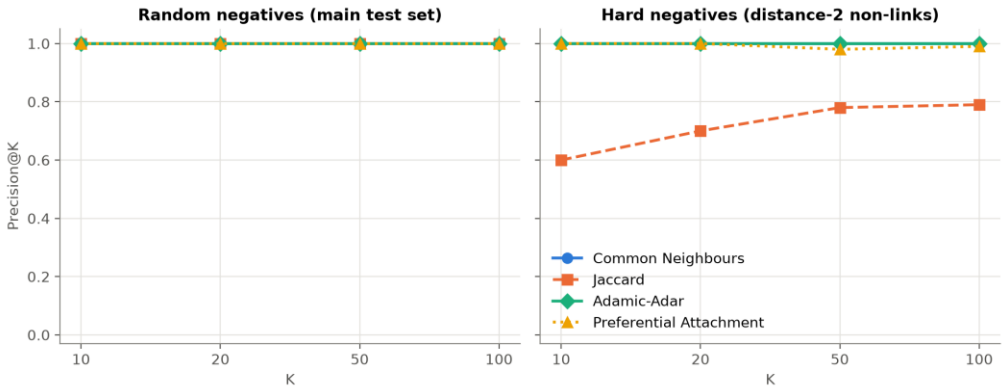
ROC and Precision-Recall Curves

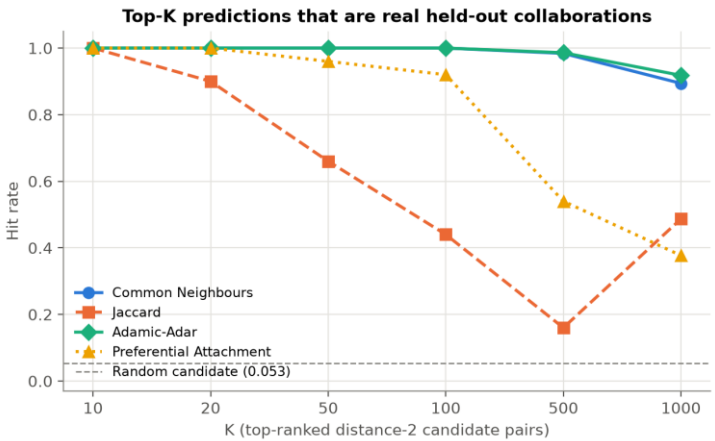




CN, Jaccard and Adamic-Adar overlap; Preferential Attachment ranks pairs far less sharply.

Top-K Comparison





Hard negatives	ROC-AUC	AP	P@10
Common Neighbours	0.778	0.814	1.000
Jaccard	0.759	0.788	0.600
Adamic-Adar	0.850	0.900	1.000
Preferential Attachment	0.644	0.681	1.000

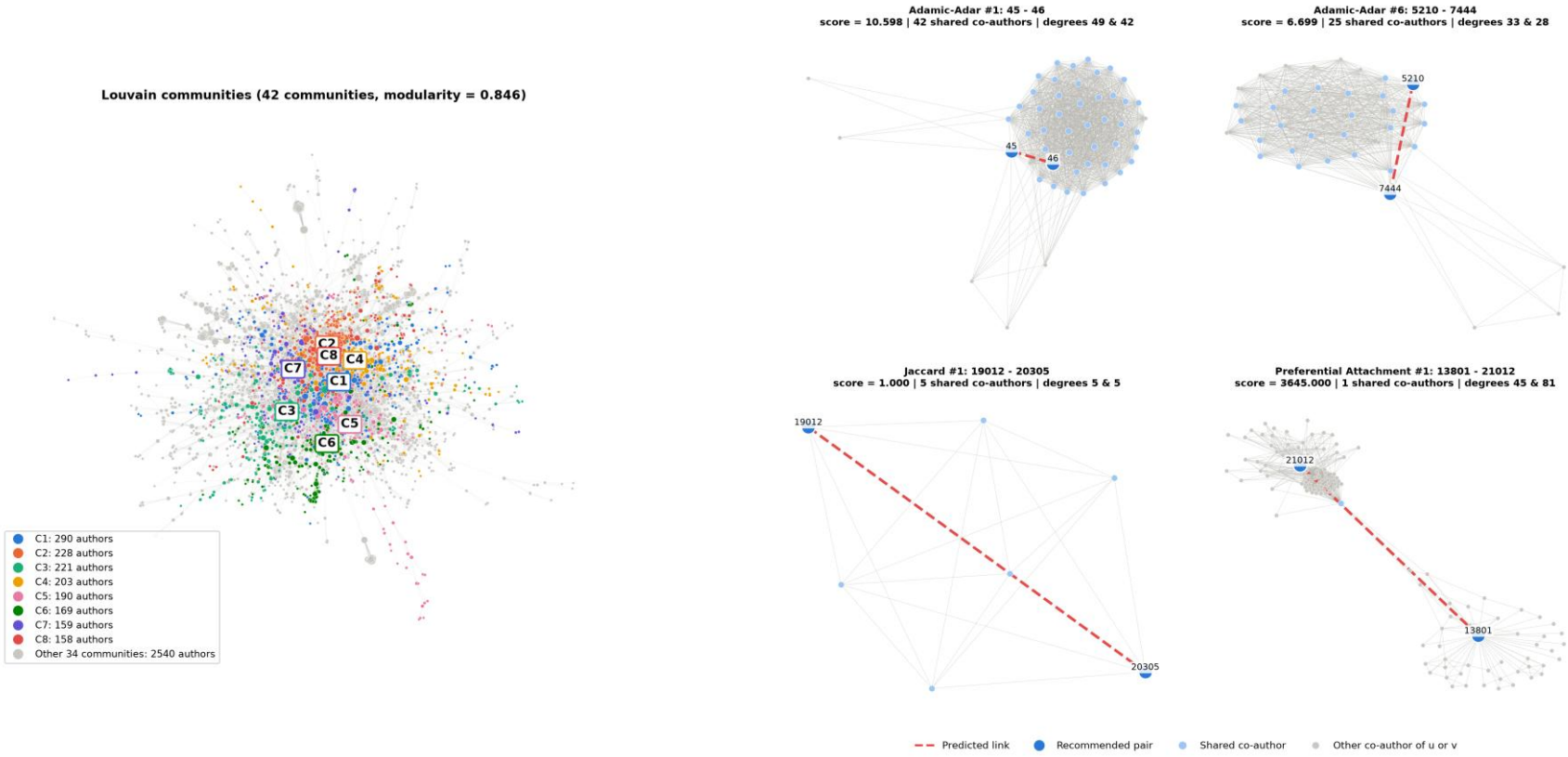
- With random negatives P@K = 1.0 for all methods (saturated).
- Hard negatives (share >= 1 co-author) separate them: Adamic-Adar best.
- Hit rate: 100% of Adamic-Adar's Top-100 (over 46,588 candidates) are real held-out links vs 5.3% at random.

Recommendations (Adamic-Adar)

#	Author u	Author v	AA score	Shared co-authors	k(u)	k(v)
1	45	46	10.598	42	49	42
2	4511	4513	10.592	42	45	43
3	4511	22691	10.592	42	45	77
4	46	4513	10.335	41	42	43
5	46	22691	10.335	41	42	77
6	5210	7444	6.699	25	33	28
7	7444	8727	6.699	25	28	33
8	7444	9869	6.699	25	28	41
9	7444	12710	6.699	25	28	33
10	7444	15245	6.699	25	28	33

- Candidates: 63,288 non-adjacent pairs at distance 2 (every pair further away scores 0 for CN/JC/AA).
- Filtered: no self-pairs, no existing links, no (v, u) duplicates.
- Top-20 pairs share 24-42 co-authors; their groups have edge density 0.997-0.999 - missing links inside near-cliques (large multi-author papers).
- Why #1? Authors 45 and 46 share 42 co-authors but have never published together.

Visualization



Left: Louvain communities of the LCC. Right: ego networks of each method's #1 pair - dashed red = predicted link, light blue = shared co-authors.

Insights

- **Clustering drives prediction:** 89% of held-out links fall inside one Louvain community - new collaborations are local.
- **Adamic-Adar wins** because it discounts shared hubs (hard-negative ROC-AUC 0.850 vs 0.778 for CN).
- **Jaccard over-normalises:** its top scores go to tiny groups with identical co-author lists (hard-negative P@10 = 0.600).
- **PA favours hubs:** it recommends popular authors from different groups, not close ones (ROC-AUC 0.849).
- **Stable:** ROC-AUC std over 5 splits <= 0.0043 for every method.

Limitations and Future Work

Limitations

- Random, not temporal split (no timestamps in SNAP data)
- Spanning-tree split favours easier (cycle) test edges
- Random negatives are easy; 1:1 hides the real imbalance
- Local indices cannot reach pairs at distance >= 3
- Large cliques dominate the Top-K; IDs are anonymised

Future work

- Katz index / Resource Allocation (longer paths)
- Node2Vec embeddings or graph neural networks
- Supervised classifier on CN, JC, AA, PA + community features
- Temporal evaluation; weighted edges (number of papers)
- Diversity-aware re-ranking of recommendations

Conclusion

- Modeled ca-GrQc as an author graph (4,158 authors, 13,422 links in the LCC): sparse, highly clustered, small-world, modular.
- Built a leakage-free link prediction pipeline with a connectivity-preserving split.
- Adamic-Adar is the best predictor: ROC-AUC 0.9568 +/- 0.0028, hard-negative ROC-AUC 0.850, 100% hit rate in its Top-100.
- Delivered explained Top-20 recommendations, visualizations, a Cytoscape export, a dataset package and a reproducible notebook.

Thank you !

Questions?